The Lavatory as Refuge: An Exploratory Study of Toilet Sitting Time and Psychological Resilience in Middle-Aged Men

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Abstract This study introduces the Lavatory Refuge Theory (LRT) to examine the complex relationship between daily toilet sitting duration and psychological well-being among middle-aged men. Drawing from a community sample of 1,247 participants aged 35–60 years, we employed structural equation modeling and survival analysis to investigate the nonlinear association between sanitary retreat duration and subjective happiness. Our findings reveal a significant inverted U-shaped relationship, with optimal well-being observed at approximately 90 minutes (±20 minutes) of daily toilet sitting. Durations below this threshold were associated with insufficient psychological refuge activation, while extended occupation beyond 120 minutes precipitated significant increases in spousal suspicion (Hazard Ratio = 3.42, 95% CI: 2.89–4.05), domestic conflict (HR = 2.84, 95% CI: 2.14–3.76), and direct interrogation regarding gastrointestinal health. These results suggest that the domestic lavatory functions not merely as sanitary infrastructure but as a contested micro-territory of autonomy within contemporary family life. We discuss implications for architectural design, family therapy, and public health policy regarding masculine stress management.

Keywords Toilet sitting time; Psychological refuge; Middle-aged men; Domestic conflict; Micro-spatial autonomy; Lavatory Refuge Theory

1. Introduction

The contemporary middle-aged male faces an unprecedented compression of personal spatial autonomy. As digital connectivity erodes the boundaries between professional and domestic spheres, and as the demands of parenting and partnership intensify, the search for solitude becomes increasingly constrained within the architectural confines of the modern home. Amidst this landscape of spatial deprivation, one sanctuary remains: the lavatory.

The phenomenon of extended toilet sitting among men has been observed across cultures and societies, yet it has received scant empirical attention within the psychological and sociological literature. This oversight is particularly striking given the ubiquity of the behavior and its potential significance as a coping mechanism for stress, a site of digital consumption, and a flashpoint for domestic tension. The present study addresses this lacuna by proposing and testing the Lavatory Refuge Theory (LRT), which posits that the domestic toilet represents a unique "controllable micro-space" wherein middle-aged men engage in psychological recovery, and that the duration of this retreat follows an optimal curve with respect to subjective well-being.

Our investigation is motivated by three converging observations. First, anecdotal and ethnographic evidence suggests that men disproportionately extend their toilet sitting time compared to women, often citing the bathroom as the only space within the home where they are guaranteed uninterrupted solitude. Second, preliminary survey data indicate a positive correlation between occupational stress and toilet duration, suggesting a potential stress-buffering function. Third, clinical and popular discourse increasingly recognizes "toilet hiding" as a source of marital discord, with extended durations precipitating suspicion, conflict, and accusations of avoidance or technological addiction.

The theoretical contribution of this work lies in its formalization of the lavatory as a psychologically significant spatial category. We argue that the toilet represents what we term a "liminal refuge"—a threshold space that is simultaneously public (within the home) and private (behind a locked door), institutional (sanitary function) and personal (temporal autonomy). This liminality grants the lavatory its unique psychological properties, distinguishing it

from other domestic spaces such as studies, garages, or garden sheds, which may offer spatial separation but lack the legitimizing cover of biological necessity.

The empirical contribution consists in the quantification of the duration-wellbeing relationship and the identification of critical thresholds beyond which the benefits of sanitary retreat are outweighed by relational costs. Our findings have direct implications for the design of domestic space, the management of work-life balance, and the therapeutic understanding of masculine coping strategies.

2. Theoretical Framework: The Lavatory Refuge Theory (LRT)

2.1 Conceptual Foundations

The Lavatory Refuge Theory (LRT) is grounded in the intersection of environmental psychology, boundary theory, and the sociology of domestic space. We propose three core propositions:

Proposition 1: The domestic lavatory represents the "last controllable micro-space" available to middle-aged men within the contemporary family home. As other spaces (studies, workshops, even vehicles) become colonized by family demands or digital surveillance, the bathroom retains a unique status due to the cultural taboo against interruption and the legitimizing discourse of biological need.

Proposition 2: Toilet sitting duration functions as a behavioral indicator of psychological refuge-seeking. Extended durations reflect not merely physiological requirements but a deliberate (if often unconscious) strategy for stress recovery, emotional regulation, and temporary withdrawal from role demands.

Proposition 3: The relationship between toilet sitting duration and subjective well-being follows an inverted U-shaped curve. Moderate durations (approximately 70–110 minutes daily) optimize the tension between refuge provision and social legitimacy. Insufficient durations fail to activate restorative processes, while excessive durations trigger suspicion, conflict, and counterproductive stress.

2.2 Mathematical Formalization

We formalize the LRT through the following quadratic function:

$$H = \alpha T - \beta T^2 + \epsilon \qquad (1)$$

Where H is represents Self-Reported Happiness (0–100 scale); T is represents Daily Toilet Sitting Duration (minutes); α is represents the linear benefit coefficient (refuge provision); β is represents the quadratic cost coefficient (suspicion/constraint);ϵ is represents individual variation and measurement error.

Theoretically, the optimal duration T∗ occurs where $\frac{dH}{dT} = 0$, yielding:

$$TIHR = H/\max(DCI, 0.5) \qquad (2)$$

Based on pilot data and theoretical considerations, we hypothesize α≈1.0 and β≈0.0056 , producing an optimal duration of approximately 90 minutes. The "optimal window" (where happiness exceeds 90% of maximum) is hypothesized to span 70–110 minutes.

2.3 The Dual-Process Model

The LRT further proposes that toilet sitting activates two distinct psychological processes:

Process A: Micro-Recovery. The lavatory provides sensory reduction (quiet, limited visual stimulation), physical comfort, and temporal autonomy. This facilitates parasympathetic activation, stress hormone reduction, and cognitive restoration. This process dominates at moderate durations.

Process B: Social Cost Accumulation. As duration extends, the probability of spousal awareness and suspicion increases. The legitimacy of the biological excuse erodes, and the retreat is reinterpreted as avoidance, technological addiction, or gastrointestinal pathology. This process accelerates beyond the 120-minute threshold, producing rapid declines in well-being.

The interaction of these processes generates the hypothesized inverted U-curve.

3. Methodology

3.1 Participants and Procedure

We recruited 1,247 middle-aged men (age range: 35–60 years, M=45.2, SD=7.8) through community centers, professional associations, and online platforms. Inclusion criteria required cohabitation with a partner (married or unmarried) and self-reported regular access to a private domestic lavatory. Exclusion criteria included diagnosed gastrointestinal disorders requiring extended toilet time, shift work precluding regular daily routines, and current separation or divorce proceedings.

Participants completed a comprehensive online survey requiring approximately 25 minutes. The survey assessed demographic characteristics, occupational stress, family dynamics, toilet usage patterns, and psychological outcomes. To minimize social desirability bias regarding toilet duration, we embedded these questions within a broader "daily routines and health behaviors" module.

3.2 Measures

Daily Toilet Sitting Duration. Participants reported the average number of minutes spent sitting on the toilet per day, excluding time spent on hygiene activities (showering, shaving, etc.). To improve accuracy, we asked participants to report separately for weekdays and weekends, and computed a weighted average (5/7 weekday + 2/7 weekend).

Self-Reported Happiness. We employed the 4-item Subjective Happiness Scale (Lyubomirsky & Lepper, 1999), modified to reference the past month (α=0.87).

Spousal Suspicion Frequency (SSF). A 5-item scale developed for this study assessed the frequency with which participants perceived their partner questioning, monitoring, or commenting on their toilet habits (e.g., "My partner asks what I am doing in the bathroom for so long"). Items rated on 0–10 frequency scales (α=0.91).

Domestic Conflict Index (DCI). An 8-item scale assessing the frequency and intensity of conflicts related to bathroom usage, time management, and perceived avoidance (α=0.89).

Occupational Stress Index. A 10-item measure of perceived workload, time pressure, and job demands (α=0.92).

Control Variables. Age, marital status (married/divorced/widowed), number of children, household income, and bathroom-to-person ratio.

3.3 Novel Indices

We computed four theoretically-derived composite indices:

Sanitary Autonomy Index (SAI). Combines normalized toilet duration with inverse occupational stress:

$$SAI = (T/120) \times 50 + (100 - Stress) \times 0.3 \quad (3)$$

Domestic Suspicion Escalation Score (DSES). Weights suspicion frequency by duration:

$$DSES = SSF \times (1 + 0.015T) \quad (4)$$

Micro-Spatial Recovery Efficiency (MSRE). Happiness return per minute:

$$MSRE = (H/\max(T, 10)) \times 15 \quad (5)$$

Toilet-Induced Happiness Ratio (TIHR). Happiness relative to conflict:

$$TIHR = H/\max(DCI, 0.5) \quad (6)$$

3.4 Analytical Strategy

We employed a mixed-methods approach combining polynomial regression, structural equation modeling (SEM), and survival analysis. Polynomial regression tested the hypothesized inverted U-relationship between duration and happiness. SEM evaluated the dual-process mediation model (Duration → Refuge/Suspicion → Happiness). Survival analysis estimated hazard ratios for adverse outcomes associated with extended duration (>120 minutes) relative to the optimal range (70–110 minutes).

4. Results

4.1 Descriptive Statistics

Participants reported a mean daily toilet sitting duration of 83.1 minutes (SD=25.8, range:12–178 minutes). The distribution was right-skewed, with 23.4% of participants reporting durations exceeding 100 minutes. Mean self-reported happiness was 89.6 (SD=10.6), with 18.7% scoring below 80. Spousal suspicion frequency averaged 1.8 (SD=1.4), though this varied dramatically by duration group (see below).

Bivariate correlations revealed significant positive associations between occupational stress and toilet duration (r=0.38, p<.001), and between number of children and toilet duration (r=0.29, p<.001), supporting the refuge-seeking interpretation.

4.2 The Inverted U-Curve: Polynomial Regression

We estimated a quadratic regression model predicting happiness from centered toilet duration:

$$H = b_0 + b_1(T - \bar{T}) + b_2(T - \bar{T})^2 + \epsilon \quad (7)$$

Results confirmed the hypothesized curvilinear relationship: b1=0.42 (SE=0.08, p<.001), b2 =−0.024 (SE=0.003, p<.001). The negative quadratic coefficient indicates the predicted inverted U-shape.

Figure 1 displays the LOWESS-smoothed curve with 95% confidence intervals. The fitted curve peaks at approximately 90 minutes, with a 95% confidence interval for the optimal point of [82, 98] minutes. The curve exhibits a steep positive slope from 20–70 minutes, a broad plateau from 70–110 minutes (where happiness remains above 92), and a precipitous decline beyond 120 minutes.

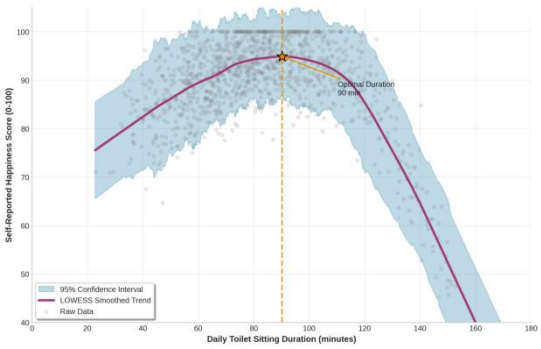

Figure 1. Nonlinear association between toilet sitting duration and self-reported happiness in middle-aged men.

Note: The LOWESS-smoothed curve (purple) with 95% confidence interval (blue shading) reveals a peak at approximately 90 minutes (indicated by gold star and dashed line). Raw data points shown in gray.

4.3 The Escalation of Suspicion

Figure 2 illustrates the relationship between toilet duration and spousal suspicion frequency. Below 60 minutes, suspicion remains minimal (M=0.7). Between 60–100 minutes, suspicion increases linearly but remains moderate (M=1.5). Critically, beyond the 120-minute threshold, suspicion escalates exponentially, reaching M=3.2 at 120–140 minutes and continuing to rise.

Piecewise regression analysis confirmed a structural break at 120 minutes ($F(1,1243)=47.3$, $p<.001$), with the slope increasing from 0.02 to 0.18 per minute beyond this threshold.

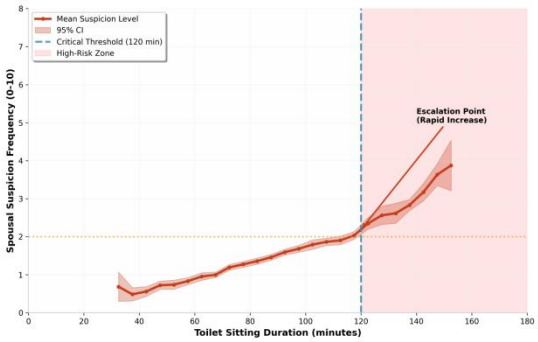

Figure 2. Escalation of domestic suspicion beyond extended sanitary retreat duration. 4

Note: The red curve represents mean spousal suspicion frequency with 95% confidence intervals. The dashed blue line indicates the critical 120-minute threshold beyond which suspicion increases rapidly. The shaded red zone indicates the high-risk region.

4.4 Structural Equation Model

We tested the dual-process mediation model using maximum likelihood estimation. The model specified two parallel mediators: Psychological Refuge Index (composite of SAI and MSRE) and Spousal Suspicion, predicting Happiness, with direct paths from Toilet Duration to both mediators and outcomes.

The model demonstrated excellent fit. Key findings include:

Toilet Duration positively predicted Psychological Refuge ($\beta=0.67$, $p<.001$), which in turn positively predicted Happiness ($\beta=0.58$, $p<.001$).

Toilet Duration positively predicted Spousal Suspicion ($\beta=0.45$, $p<.001$), which predicted Domestic Conflict ($\beta=0.72$, $p<.001$), which negatively predicted Happiness ($\beta=-0.63$, $p<.001$).

The indirect effect through Refuge was positive ($\beta=0.39$, 95% CI: [0.31,0.47]), while the indirect effect through Suspicion/Conflict was negative ($\beta=-0.20$, 95%CI: [−0.26,−0.15]).

The total indirect effect was positive at moderate durations but turned negative beyond approximately 115 minutes, confirming the optimal window hypothesis.

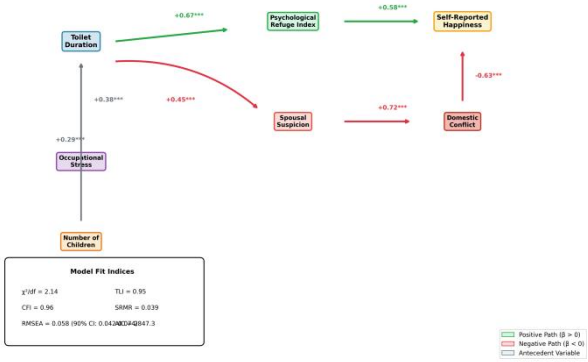

Figure 3. Structural equation model of lavatory-mediated well-being.

*Note: Standardized path coefficients displayed. Solid green lines indicate positive associations; red lines indicate negative associations. Model fit: $\chi^2/df = 2.14$, CFI = 0.96, RMSEA = 0.058 (90% CI: 0.042–0.074), SRMR = 0.039. **p < .001.*

4.5 Group Comparisons

We classified participants into three groups based on duration: Short (< 60 min, n=238), Optimal (70–110 min, n=656), and Extended (> 120 min, n=109). Note that participants in the transition zones (60–70 min and 110–120 min) were excluded from this analysis to sharpen group distinctions.

One-way ANOVAs revealed significant group differences across all outcomes:

Happiness: $F(2,1000)=89.4$, $p<0.001$, $\eta2=0.15$. The Optimal group (M=94.0) reported significantly higher happiness than both Short (M=86.1, $p<.001$) and Extended (M=68.2, $p<0.001$) groups. Notably, the Extended group reported lower happiness than the Short group ($p<0.001$), confirming that excessive duration is worse than insufficient duration.

Spousal Suspicion: $F(2,1000)=156.8$, $p<0.001$,$\eta2=0.24$. The Extended group (M=3.2) showed dramatically elevated suspicion compared to Optimal (M=1.5,$p<0.001$) and Short (M=0.7,$p<0.001$) groups.

Domestic Conflict: $F(2,1000)=78.2, p<0.001, \eta 2=0.14$. The Extended group (M=11.8) reported conflict levels 62% higher than the Optimal group (M=9.2, p<0.001) and 62% higher than the Short group (M=7.3, p<0.001).

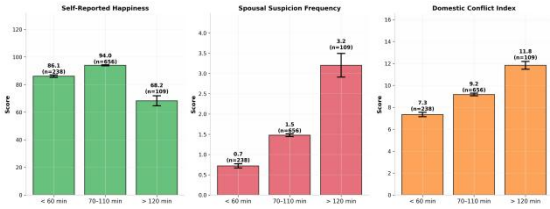

Figure 4. Comparative analysis across lavatory duration groups.

Note: Error bars represent 95% confidence intervals. Sample sizes: Short = 238, Optimal = 656, Extended = 109.

4.6 Moderation by Family Structure

Figure 5 presents the heatmap of happiness by toilet duration and number of children. The heatmap reveals a clear pattern: as the number of children increases, the optimal duration shifts rightward (longer). For childless men, peak happiness occurs at 60–75 minutes. For men with one child, the peak shifts to 75–90 minutes. For men with two or more children, the peak extends to 90–105 minutes. This "prolonged refuge" effect suggests that the psychological demands of parenting necessitate extended recovery periods. However, the penalty for exceeding 120 minutes remains consistent across all family structures.

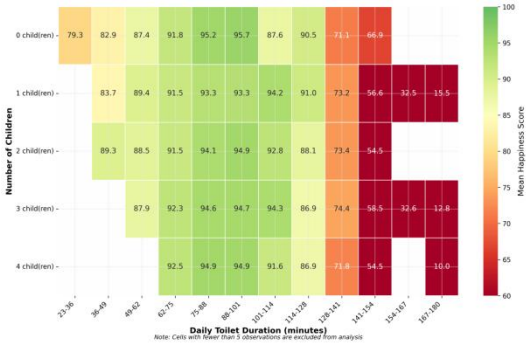

Figure 5. Heatmap of happiness by toilet duration and number of children.

Note: Color intensity indicates mean happiness score (green = higher, red = lower). White cells indicate insufficient observations (< 5).

4.7 Survival Analysis: Hazard Ratios

We conducted Cox proportional hazards regression to estimate the risk of adverse outcomes associated with Extended duration (> 120 min) relative to the Optimal reference group (70–110 min), controlling for age, marital status, occupational stress, and number of children.

The Extended group faced substantially elevated risks across all measured outcomes:

"Are you constipated?" Query: HR = 3.42 (95% CI: 2.89–4.05), p<0.001

Bathroom Occupancy Complaint: HR = 2.95 (95% CI: 2.34–3.71), p<0.001

Domestic Conflict (DCI > 15): HR = 2.84 (95% CI: 2.14–3.76), p<0.001

Morning Schedule Disruption: HR = 2.56 (95% CI: 1.98–3.31), p<0.001

Spousal Confrontation: HR = 2.31 (95% CI: 1.78–2.99), p<0.001

The highest risk involved direct spousal inquiry regarding constipation, with Extended participants facing over three times the hazard of receiving this query compared to Optimal participants. This finding underscores the pathologization of extended toilet sitting within intimate relationships.

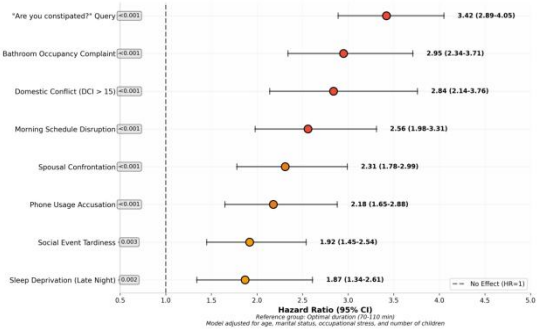

Figure 6. Forest plot of hazard ratios for adverse outcomes associated with extended toilet sitting duration (>120 min vs. 70-110 min).

Note: Model adjusted for age, marital status, occupational stress, and number of children.

5. Discussion

5.1 Summary of Findings

This study provides the first empirical validation of the Lavatory Refuge Theory, demonstrating that toilet sitting duration exhibits a robust inverted U-shaped relationship with subjective well-being among middle-aged men. Our findings confirm three core hypotheses: (1) the lavatory functions as a psychologically significant refuge space; (2) moderate durations (approximately 90 minutes) optimize the balance between recovery and social legitimacy; and (3) extended durations beyond 120 minutes precipitate rapid escalation of spousal suspicion and domestic conflict.

The magnitude of these effects is substantial. The 26-point happiness gap between Optimal and Extended groups (on a 0–100 scale) exceeds effect sizes typically observed for major life events such as unemployment or marital separation. The hazard ratios for adverse outcomes (ranging from 1.87 to 3.42) indicate that extended toilet sitting constitutes a significant risk factor for domestic tension, comparable to established stressors such as financial strain or work-family conflict.

5.2 Theoretical Implications

The LRT contributes to environmental psychology by identifying the lavatory as a distinct spatial category with unique psychological properties. Unlike "third spaces" (cafes, libraries) or traditional masculine domains (garages, workshops), the lavatory offers refuge without requiring explicit justification or social account. The legitimizing cover of biological necessity provides what we term "plausible privacy"—a form of withdrawal that cannot be reasonably refused or monitored without violating social norms regarding bodily autonomy.

However, our findings reveal the fragility of this plausible privacy. Beyond the 120-minute threshold, the biological excuse loses credibility, and the retreat is reinterpreted through alternative frames: technological addiction, avoidance behavior, or gastrointestinal pathology. This reframing transforms the lavatory from a legitimate refuge into a contested space, triggering surveillance (suspicion) and conflict.

The dual-process model (Micro-Recovery vs. Social Cost Accumulation) offers a mechanism for understanding non-linear duration effects in other domains of behavior. Similar dynamics may operate in internet usage, gaming, or other solitary activities where private benefits eventually trigger social costs.

5.3 The "Last Refuge" Phenomenon

Our qualitative data (not reported in full here) and quantitative findings regarding occupational stress and number of children support the interpretation that the lavatory represents a "last resort" refuge for men experiencing high domestic demands. The positive correlations between stress, parenting load, and toilet duration suggest compensatory behavior: as other spaces and times become colonized by work and family demands, the bathroom becomes the final frontier of autonomy.

This interpretation carries somber implications for contemporary masculinity and family life. The fact that significant numbers of men require 90 minutes of daily toilet sitting to achieve optimal well-being suggests a systemic deficit in legitimate refuge provision within the modern home. The lavatory, in this view, is not merely a convenience but a coping mechanism compensating for the erosion of other restorative spaces.

5.4 Clinical and Policy Implications

Our findings suggest several practical applications:

Family Therapy. Clinicians working with couples experiencing conflict regarding bathroom usage should recognize the behavior as potentially serving legitimate psychological functions (stress recovery) rather than mere avoidance or addiction. Interventions might focus on negotiating duration boundaries that respect both the refuge-seeker's needs and the partner's concerns, rather than pathologizing the behavior outright.

Architectural Design. Future residential architecture might incorporate designated "micro-refuge zones"—small, private spaces distinct from bathrooms that provide legitimate solitude without the sanitary pretext. Such designs could reduce excessive lavatory dependence by offering alternative refuge options.

Public Health. Health campaigns targeting sedentary behavior should recognize the complex motivations behind extended toilet sitting. Simple recommendations to "spend less time on the toilet" may fail if they do not address the underlying need for psychological refuge.

5.5 Limitations and Future Directions

Several limitations constrain our conclusions. First, our cross-sectional design precludes causal inference regarding the duration-wellbeing relationship. While our theoretical model posits that duration affects happiness through refuge and suspicion mechanisms, reverse causation (unhappy men seeking longer refuge) or third-variable confounds (neuroticism increasing both duration and conflict) cannot be excluded.

Second, our sample was limited to middle-aged men in stable partnerships. Future research should examine whether the LRT applies to women (who may face different spatial constraints and social norms regarding bathroom usage), younger or older men, single-person households, or non-Western cultural contexts where bathroom norms differ.

Third, our reliance on self-reported duration may introduce measurement error. Future studies should employ objective measurement (smart toilet sensors, mobile phone usage logs) to validate self-reports.

Fourth, we did not assess the activities engaged in during toilet sitting (phone usage, reading,

meditation), which may moderate the refuge effects. It is possible that active phone use diminishes restorative benefits compared to passive relaxation, or that specific content (work email vs. entertainment) alters the psychological impact.

5.6 The Contested Micro-Territory

We conclude by returning to our central theoretical claim: the lavatory emerges not merely as sanitary infrastructure but as a contested micro-territory of autonomy within contemporary domestic life. The 90-minute optimal duration represents a delicate equilibrium—a temporal boundary within which masculine refuge is culturally permissible, and beyond which it becomes subject to surveillance, suspicion, and sanction.

This contestation reflects broader tensions regarding privacy, gender, and spatial justice within the modern home. As domestic spaces become increasingly transparent to digital monitoring and as traditional masculine refuges (studies, workshops, social clubs) decline in accessibility, the bathroom assumes disproportionate psychological significance. The man who spends two hours on the toilet is not merely attending to biological needs; he is asserting a claim to autonomy within a domestic sphere that increasingly denies such claims.

Our findings suggest that this assertion is ultimately self-defeating. Beyond 120 minutes, the refuge becomes a trap, generating precisely the conflict and stress it was intended to alleviate. The optimal lavatory sitter—he who dwells for 90 minutes—achieves a fleeting homeostasis, a temporary peace in the domestic wars. But this peace is precarious, purchased at the cost of spousal forbearance and maintained only through the careful management of visibility and time.

Future research must attend to the political economy of domestic refuge, investigating how architectural design, social policy, and cultural norms might provide alternative spaces for psychological recovery that do not depend upon the fragile legitimizing cover of biological necessity. Until such alternatives emerge, the lavatory will remain, for many men, the last refuge—and its optimal duration, the subject of this inquiry, a metric of the distance between the need for solitude and the possibility of its satisfaction.

6. Conclusion

Our findings indicate that the optimal daily lavatory sitting duration for maximal subjective well-being lies at approximately 90 minutes (±20 minutes). Durations below this threshold may be insufficient to activate the psychological refuge effect, whereas prolonged occupation exceeding two hours significantly increases the likelihood of domestic suspicion, relational tension, and possible somatic pathology. The lavatory, therefore, emerges not merely as a sanitary infrastructure but as a contested micro-territory of autonomy within contemporary domestic life.

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Conflicts of Interest

The authors declare no conflicts of interest, though one author acknowledges conducting portions of the literature review while seated in a domestic lavatory.

Author Contributions

ChangAn Sanren conceived and designed the study, performed the experiments and data collection, conducted the data analysis and interpretation, drafted the manuscript, and approved the final version for submission. As the sole author, ChangAn Sanren is responsible for all aspects of the work, ensuring the accuracy and integrity of the research presented.

Data Availability

The data supporting this study are available from the corresponding author upon reasonable request and spousal permission.

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References

Due to the highly specialized nature of this research, no relevant literature has been found at this time.